Product Requirements Document (PRD)

Product overview

This product is a student-focused rental marketplace for finding PGs, flats, hostels, single rooms, co-living spaces, and small rental properties near selected colleges. The V1 is focused on helping students quickly discover nearby housing and helping owners list verified properties with admin approval before listings go live.

The product will launch as a usable V1 with a web frontend built using Vite + React + Tailwind, a React Native mobile app, a FastAPI backend, Firebase Authentication, Firestore, Cloudinary for images, and OpenStreetMap + Leaflet for maps.[cite:29] [cite:31] [cite:37]

Problem statement

Students searching for housing near colleges often rely on WhatsApp groups, brokers, scattered listings, and incomplete local references. This makes it hard to compare options by distance, rent, room sharing type, amenities, and trustworthiness.

Property owners also face a fragmented process. They do not have a simple way to list college-nearby properties, manage inquiries, update availability, and reach verified student demand in one place.

The product solves this by creating a college-first marketplace where users choose a college first, then discover rental inventory within a defined radius with structured listing information and direct owner contact.

Goals

- Help students find nearby rental options faster.
- Help owners publish and manage listings without broker dependency.
- Give admins full control over listing quality, fraud prevention, college master data, and platform moderation.
- Launch a focused V1 without building unnecessary communication or financial tools.

Non-goals

The following are explicitly out of scope for V1:

- In-app chat.
- Rent payment collection.
- Digital rent agreement generation.
- Roommate matching.

- Referral or reward programs.
- Advanced recommendation engine.
- Complex brokerage workflows.
- Multi-city expansion workflows beyond the initial college-led launch structure.
- Full property management software for landlords.

Target users

Students / tenants

Students want affordable, nearby, trustworthy accommodation near their college. They care most about distance from college, rent, sharing type, amenities, food availability, and direct contact with the owner.

Owners

Owners include PG operators, flat owners, hostel managers, and single-room landlords. They want a fast way to create listings, upload photos, update availability, and receive inquiries from relevant students.

Admins

Admins need a complete control panel to approve listings, manage colleges, moderate users, handle spam, and monitor platform health. Admin is a high-trust internal role and must have full visibility across students, owners, and listings.

User roles

There will be three login roles in V1:

- Admin
- Student
- Owner

A single Firebase Authentication system will be used for login, and role access will be enforced in the backend and app logic using role data stored against the user profile.[cite:31][cite:37]

Core user flows

Student flow

1. Open app or website.
2. Select college.
3. Apply radius, property type, budget, gender, and amenity filters.

4. Browse listing cards sorted by nearest, price, newest, or rating.
5. Open listing details.
6. Review photos, room options, rent, deposit, rules, amenities, food details, availability, reviews, and map location.
7. Call or WhatsApp owner directly.
8. Optionally log in to shortlist listings, save alerts, submit inquiries, and leave reviews after move-in.

Owner flow

1. Log in using phone OTP or approved authentication method.
2. Complete owner profile and property identity details.
3. Create new listing.
4. Choose property type.
5. Fill dynamic form fields based on the selected property type.
6. Upload photos.
7. Set map pin and address.
8. Submit listing for review.
9. Track inquiries and listing status from dashboard.
10. Edit listing, mark occupied, or mark available.

Admin flow

1. Log in as admin.
2. Review pending owner and listing submissions.
3. Approve, reject, or flag listings.
4. Manage colleges and searchable college metadata.
5. View users by role.
6. Mark listings as featured.
7. Remove spam or fraudulent content.
8. Review platform analytics and operational metrics.

V1 feature list

Student-facing MVP features

- College selector as the first search control.
- Radius filter from selected college.
- Property type filter for PG, Flat, Shop to Rent, Hostel, Single Room, and Co-living.

- Gender filter for relevant property types.
- Budget range filter.
- Amenity filters including Food, WiFi, AC, Parking, Laundry, Geyser, and CCTV.
- Sort by price, distance, newest, and rating.
- Listing cards with essential summary details.
- Listing detail page with photos, distance, room options, rent, deposit, amenities, food details, owner contact, map, reviews, and availability.
- Call owner and WhatsApp owner actions.
- Student login.
- Shortlist / save listing.
- Recently viewed listings.
- Inquiry form or action from listing page.
- Listing alerts based on saved search conditions.
- Review submission after move-in or verified stay policy.

Owner-facing MVP features

- Owner login.
- Owner profile setup.
- Dynamic listing form based on property type.
- Photo upload with compression through Cloudinary.
- Location pin selection using map tools.
- Listing submit, edit, and delete.
- Mark listing as Available or Occupied.
- Dashboard with views and inquiries.

Admin MVP features

- Admin login.
- Dashboard overview.
- Approve / reject listings.
- Manage all users.
- Manage all listings.
- Add, edit, and disable colleges.
- Mark listing as featured.
- Remove spam or fake listings.
- View analytics including top colleges, top searched areas, listing views, and inquiry volume.

Future features

These should be planned for later, not built in V1:

- Map-first search page with all live listing pins.
- In-app chat.
- Digital rent agreement PDF generator.
- Roommate finder.
- Referral program.
- Paid owner plans and billing automation.
- Advanced recommendations and ranking personalization.
- Rich owner CRM and follow-up workflows.

Functional requirements

Search and discovery

- User must select a college before distance-based browsing begins.
- Radius filter must support 0.5 km, 1 km, 2 km, and 5 km options.
- Listings must display computed distance from the selected college.
- Search should support filtering by category, budget, amenities, and gender where relevant.
- Sorting must be available by nearest, lowest price, newest, and highest rated.

Listing details

Each listing should support:

- 3 to 5 swipeable images at minimum display level.
- Property name.
- Property type badge.
- Distance from selected college.
- Room options and pricing table.
- Monthly rent and security deposit.
- Full amenities list.
- Food menu or food availability details where relevant.
- Owner name and contact methods.
- Map location.
- Ratings and reviews.
- Availability status.

- House rules.

Property-specific fields

PG

- Gender.
- Food included or not.
- Mess timing.
- Gate timing.
- Warden contact.

Flat

- BHK type.
- Furnished / semi-furnished / unfurnished.
- Floor.
- Lift availability.
- Society name.

Shop

- Area in sq ft.
- Floor.
- Allowed use type.
- Maintenance charge.

Hostel

- Total capacity.
- Study hall availability.
- Curfew time.
- Warden on-site status.

Single Room

- Attached or shared bathroom.
- Kitchen access.
- Furnished status.

Co-living

- Community events.
- Co-working availability.
- Minimum stay duration.

Authentication and roles

- Student and owner access must use Firebase Authentication.[cite:30] [cite:31]
- Admin login must be protected and restricted to approved admin accounts.[cite:31]
- Role checks must be enforced in backend APIs.[cite:31] [cite:34]
- Logged-out users may browse public listings and contact owners directly.
- Logged-in users are required for shortlisting, alerts, reviews, and owner dashboards.

Analytics

Free analytics must track at minimum:

- Daily active users.
- Listing views.
- College searches.
- Inquiry actions.
- Shortlist actions.
- Owner listing submissions.
- Admin approvals and rejections.

Firebase Analytics is a practical free fit for this stack, especially because it integrates with Firebase-backed apps and event tracking workflows.[cite:30] [cite:36]

Edge cases

- No listings found for selected college and radius.
- College exists but no active inventory in chosen property type.
- Listing marked available but all room options filled.
- Duplicate owner accounts using same phone number.
- Owner starts a listing but leaves required fields incomplete.
- Uploaded images fail or exceed file limits.
- Map pin placed far away from typed address.
- College coordinates missing or incorrect, causing wrong distance calculation.
- Reviews submitted by users who never stayed there.
- Owner deletes listing after receiving shortlists or inquiries.

- Spam or misleading listings created by fake owners.
- Properties with missing deposit, room pricing, or amenity data.
- One property matching multiple property-type interpretations.
- Student tries to shortlist or review without login.
- Admin rejects a listing that is already visible due to sync delay.
- Owner changes property type after entering type-specific fields.

Usability requirements

- College selection must be obvious and be the starting point of discovery.
- Search and filter interactions must be usable on both mobile web and app.
- Listing pages must prioritize rent, distance, room type, and contact actions above less critical details.
- Contact owner actions should be available without forcing login.
- Forms for owners should show only relevant fields for the selected property type.
- Admin workflows should reduce manual effort for approval and moderation.

Data and moderation requirements

- Every owner-submitted listing must remain in pending review until admin approval.
- Listings must have approval status and visibility status as separate fields.
- Colleges must be manageable by admin from a master data panel.
- User role, user status, and listing moderation logs should be stored for operational traceability.
- Images should be stored via Cloudinary to reduce asset load and support compressed delivery.

Success metrics

Adoption metrics

- Number of active listings per college.
- Number of active student users.
- Number of active owners.
- Search-to-listing-view conversion rate.

Engagement metrics

- Average listing views per active user.
- Shortlist rate.
- Inquiry rate per listing view.
- Return visits from students.

Supply quality metrics

- Listing approval rate.
- Percentage of listings with complete data.
- Percentage of listings with at least 3 valid photos.
- Median time from owner submission to admin approval.

Marketplace health metrics

- Number of available listings by college.
- Student inquiry to owner response rate.
- Fake or spam listing removal rate.
- Percentage of stale listings older than a defined freshness threshold.

V1 release scope summary

V1 should focus on one clear promise: students can find nearby rental options around a selected college and directly contact owners, while owners can list and manage inventory under admin moderation. This means V1 should prioritize searchable inventory, clean listing pages, owner submission tools, and a complete admin panel, while postponing chat, rent agreements, referral systems, and other expansion features.

Open product decisions

These decisions should be finalized before development begins:

- Whether student login supports both Google and phone OTP in V1, or only one method first.
- Whether owner registration requires manual verification before first listing.
- Whether reviews require admin approval.
- Whether alerts are email-based, app-based, or dashboard-only in V1.
- Whether shops should remain in V1 if the first launch is strictly student housing focused.